

Review retrospective:  
*the bounded-prior quartic partition — a clean tail-bound pass,  
two dead opens, and two locals the build keeps*

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### Abstract

A soundness review of `Laplace/OneD/QuarticBoundedPrior.lean`: the bounded-interval quartic partition function, with its boundary correction quantified by a Gaussian-comparison tail bound. This is the grammar paper’s  $Z_n[\varphi]$  precursor with only the bounded-prior ingredient added. Result: **a clean fidelity pass with two dead opens removed** — and two textually-idle locals deliberately left in place because the build (and an independent reviewer) flag them as implicitly consumed.

## 1 Setting

The first review of the bounded-prior / grammar-precursor strand (a sibling to the harmonic 1D stack swept earlier in this loop). The file refines the pure-quartic partition  $\int_{\mathbb{R}} e^{-tw^4/24} dw$  by restricting to  $[-a, a]$  and bounding the error:

$$\left| \int_{[-a, a]} e^{-tw^4/24} dw - \frac{1}{2} \left( \frac{24}{t} \right)^{1/4} \Gamma\left(\frac{1}{4}\right) \right| \leq \frac{24}{ta^3} e^{-ta^4/24}.$$

It leans on the Gaussian tail `TailBound` (`gaussian_tail_bound_rescaled_Ioi`, reviewed earlier in the loop).

## 2 What was reviewed

The half-line tail `quartic_tail_Ioi`; the even-symmetry tail equality `quartic_integral_Iio_neg_eq_integral_Ioi`; the three-piece split `quartic_partition_split` ( $\int_{\mathbb{R}} = \int_{[-a, a]} + 2\int_{(a, \infty)}$ ); and the headline `quartic_partition_bounded_prior`. The informal ground truth was the file’s docstrings plus the closed form  $\int e^{-cw^4} = \frac{1}{2}c^{-1/4}\Gamma(\frac{1}{4})$  ( $c = t/24$ ).

## 3 Fidelity / soundness findings

**Sound; no defect.** The tail argument is the clean comparison  $w^4 \geq a^2w^2$  for  $w \geq a$ , so  $e^{-tw^4/24} \leq e^{-(bw^2)/2}$  with  $b = ta^2/12$  (since  $b/2 = ta^2/24$ ); Mill’s ratio on the Gaussian then gives  $\int_{(a, \infty)} e^{-(b/2)w^2} \leq e^{-(ba^2)/2}/(ba) = e^{-ta^4/24} \cdot 12/(ta^3)$ . Evenness equates the two tails, the split assembles  $\int_{\mathbb{R}}$ , and the headline doubles the tail ( $12 \rightarrow 24$ ). The reviewer tabulated every constant — the comparator rate  $b = ta^2/12$ , the Mill’s constant  $12/(ta^3)$ , the doubling to  $24/(ta^3)$ , the

leading  $\frac{1}{2}(24/t)^{1/4}\Gamma(\frac{1}{4})$ , and the exponent argument  $ta^4/24$  — and found “no fidelity mismatches.” The symmetry lemma is stated without positivity hypotheses (faithful on the intended  $t, a > 0$  regime).

## 4 Simplifications applied

**Two dead opens.** `open Real` carried nothing (every `Real`-name is qualified; bare `exp/Γ` live only in comments) and `open scoped Nat` carried nothing (no `!!` notation; the only `Nat` names, `Nat.cast_zero` and `Nat.mul_zero`, are qualified). `open MeasureTheory` (bare `IntegrableOn`, `setIntegral_*`, `measurableSet_*`, `volume`) and `open Set` (bare `Ioi/Iio/ Icc/Iic`) stay. Build green, no in-file warnings, signatures byte-identical; the reviewer affirmed.

## 5 Disagreements and open questions

None. Two locals look textually unused but were correctly *not* removed: `ha` in `quartic_partition_split` is consumed implicitly by a `linarith` inside the `hunion` set-equality proof, and `htail_nn` in the headline is consumed implicitly by the `positivity` call discharging the side goal  $0 \leq 2 \cdot (\dots)$  inside `abs_of_nonneg`. The `unusedVariables` linter is silent on both, consistent with genuine (implicit) use; the reviewer independently flagged both as “not a safe dead-local deletion.” An optional refactor making the nonnegativity explicit (`mul_nonneg` (by `norm_num`) `htail_nn`) was declined as shape-only.

## 6 Lessons

- **A local consumed by `linarith/positivity` is invisible to a textual scan but is not dead.** `linarith` pulls every in-scope inequality hypothesis, and `positivity` can discharge a goal using a context fact it never names. The `unusedVariables` linter staying silent (no warning at baseline) is the positive signal that such a local is load-bearing — the absence of a warning is evidence, not just the absence of evidence. Removing on a textual “looks unused” alone would have broken the proof.
- **Pure-tail-bound files concentrate all their risk in the constants.** For a one-headline tail-estimate file the only place an error can hide is an arithmetic constant or an inequality direction; the reviewer’s value was re-deriving the comparator rate and the Mill’s constant, not re-checking the (obvious) comparison structure.

## 7 Follow-ups

- The sibling `QuarticBoundedPriorTestFn` (the  $\varphi$ -weighted extension), and the large `IntegralRemainder`, remain the last never-reviewed laplace `OneD` files.